

Quiz 7

STAT 109: Introductory Biostatistics

Quiz 7

Quiz 7 Practice Problems

These practice problems cover **Lecture 17**: Type I and Type II errors, power, and balancing errors in hypothesis testing. You should be able to state null and alternative hypotheses, define Type I and Type II errors in context, interpret decisions when a p-value and significance level are given, and know how to reduce each type of error.

Instructions

- Work **without notes** first; then check your answers against the solutions in Canvas.
 - The in-person quiz in lab will be created from these practice problems.
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Part A: Definitions

A1. Significance level

Define the **significance level** (often denoted α) in the context of hypothesis testing.

A2. Type I error

Define a **Type I error** in the context of hypothesis testing.

A3. Type II error

Define a **Type II error** in the context of hypothesis testing.

Part B: State hypotheses and define Type I and Type II errors in context

For each scenario below: (1) **Write the null hypothesis** H_0 and alternative hypothesis H_a (in symbols and/or words, as appropriate). (2) **In context, what would a Type I error be?** (3) **In context, what would a Type II error be?**

B1. Lucky coin

We want to know if a coin lands on heads more often than not (a “lucky” coin for heads). Let p = probability of heads on a single flip.

1. State H_0 and H_a .
2. What would a **Type I error** be in this context?

3. What would a **Type II error** be in this context?

B2. Lucky die

We want to know if a die lands on the face “6” more often than it should. Let p = probability of a 6 on a single roll.

1. State H_0 and H_a .
 2. What would a **Type I error** be in this context?
 3. What would a **Type II error** be in this context?
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B3. Medical treatment effectiveness

A new medical treatment is studied. We want to know if it is effective in **more than 10%** of cases. Let p = proportion of patients for whom the treatment is effective.

1. State H_0 and H_a .
 2. What would a **Type I error** be in this context?
 3. What would a **Type II error** be in this context?
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B4. Left-handed students

We want to know if the proportion of left-handed students at our school is **higher than** the national rate of 10%. Let p = proportion of students at our school who are left-handed.

1. State H_0 and H_a .
 2. What would a **Type I error** be in this context?
 3. What would a **Type II error** be in this context?
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B5. Defective items from a supplier

We want to know if the proportion of defective items from a new supplier is **less than 15%**. Let p = proportion of items from this supplier that are defective.

1. State H_0 and H_a .
 2. What would a **Type I error** be in this context?
 3. What would a **Type II error** be in this context?
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Part C: Given hypotheses, data, and p-value — could Type I or Type II be made?

For each problem, the **null and alternative hypotheses** and **data** are given. A **binomial test** was performed and the **p-value** is reported. The **cutoff for significance** is $\alpha = 0.05$. Answer: (1) **Do we reject or fail to reject H_0 ?** (2) **Could a Type I error be being made?** (Explain briefly.) (3) **Could a Type II error be being made?** (Explain briefly.) (4) **In context, what would the Type I error (or Type II error, whichever is possible) be?**

C1. Rejecting the null

Research question: Is the coin biased toward heads?

Hypotheses: $H_0 : p = 0.5$ vs $H_a : p > 0.5$, where p = probability of heads.

Data: We flip the coin 100 times and observe 62 heads.

P-value from binomial test: 0.0096.

1. Do we reject or fail to reject H_0 at $\alpha = 0.05$?
 2. Could a **Type I error** be being made? Why or why not?
 3. Could a **Type II error** be being made? Why or why not?
 4. In context, what would the **Type I error** be (i.e., what mistaken conclusion would we have made)?
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C2. Failing to reject the null

Research question: Is the proportion of left-handed students at our school higher than the national rate of 10%?

Hypotheses: $H_0 : p = 0.10$ vs $H_a : p > 0.10$, where p = proportion of students at our school who are left-handed.

Data: In a random sample of 25 students, 4 are left-handed.

P-value from binomial test: 0.21.

1. Do we reject or fail to reject H_0 at $\alpha = 0.05$?
 2. Could a **Type I error** be being made? Why or why not?
 3. Could a **Type II error** be being made? Why or why not?
 4. In context, what would the **Type II error** be (i.e., what mistaken conclusion would we have made)?
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C3. Another decision

Research question: Is the proportion of defective items from the new supplier less than 15%?

Hypotheses: $H_0 : p = 0.15$ vs $H_a : p < 0.15$, where p = proportion of items from this supplier that are defective.

Data: In a random sample of 40 items, 3 are defective.

P-value from binomial test: 0.034.

1. Do we reject or fail to reject H_0 at $\alpha = 0.05$?
 2. Could a **Type I error** be being made? Why or why not?
 3. Could a **Type II error** be being made? Why or why not?
 4. In context, what would the **Type I error** be?
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Part D: Multiple choice — reducing Type I and Type II error

D1. Reducing Type I error

Suppose we want to **reduce the probability of a Type I error** (rejecting the null when it is actually true). Which of the following would help?

- A. Use a **larger** significance level (e.g. $\alpha = 0.10$ instead of 0.05).
- B. Use a **smaller** significance level (e.g. $\alpha = 0.01$ instead of 0.05).
- C. Increase the sample size.
- D. Decrease the sample size.

Answer: _____

D2. Reducing Type II error (when α is fixed)

Assume the **Type I error rate** (significance level α) has been set and we do not change it. What can we do to **reduce the Type II error rate** (i.e., increase power)?

- A. Use a **smaller** significance level (e.g. $\alpha = 0.01$ instead of 0.05).
- B. Use a **larger** significance level (e.g. $\alpha = 0.10$ instead of 0.05).
- C. **Increase** the sample size.
- D. Decrease the sample size.

Answer: _____

End of Quiz 7 practice problems.